

Memory-Efficient Intrinsic Gating Adaptation for Enhanced On-Device Epilepsy Diagnosis

Shanjin Li, Di Wu , Shiqi Zhao , *Member, IEEE*, Jie Yang , *Senior Member, IEEE*,
and Mohamad Sawan , *Life Fellow, IEEE*

Abstract—Recently, advances in neuroscience and the rise of artificial intelligence have significantly enhanced the capabilities of epilepsy diagnosis. While EEG-based diagnosis offer a promising avenue for detecting and predicting seizure activity, practical implementation in real-world scenarios remains hindered by the heterogeneity of epilepsy and the variability of patient-specific biomarkers over time. Conventional deep learning models, trained on historical EEG, often fail to adapt to such biomarker variations, leading to degraded performance. Moreover, the computational and memory constraints of edge devices further exacerbate the challenge of on-device learning. To address these challenges, we introduce a novel framework, Memory-Efficient Intrinsic Gating Adaptation (MEIGA), designed to enhance real-world epilepsy diagnosis on resource-constrained edge devices. Our approach pre-trains a model using historical EEG data and employs lightweight adapter networks for efficient on-device tuning across new sessions, addressing session-to-session variability. By leveraging Direct Feedback Alignment (DFA), MEIGA reduces memory usage and computational overhead while maintaining high classification accuracy. Extensive experiments on the CHB-MIT epilepsy dataset demonstrate that MEIGA outperforms the pretrained-only Vision Transformer baseline, raising seizure prediction accuracy from 47.88% to 86.77% with only 3,908 tunable parameters (5.05% of the backbone). For seizure detection, MEIGA improves accuracy from 85.06% to 96.29% by adapting 2,008 parameters (17.40% of the base architecture). Further experiments on the AES dataset demonstrate that MEIGA consistently delivers strong performance across subjects and scales effectively to larger networks.

Index Terms—Memory efficient adaptation, EEG, epileptic seizure classification, direct feedback alignment, on-device learning.

I. INTRODUCTION

EPILEPSY is a chronic, non-communicable neurological disorder affecting approximately 60 million people worldwide [1]. It is characterized by recurrent and unpredictable seizures that can impact individuals of all ages, with the most common occurrences seen in children and the elderly. Despite the development of over a dozen new antiepileptic drugs in the past 15 years, about one-third of epilepsy patients remain refractory to pharmacological treatment, classifying them as drug-resistant. EEG-based diagnosis of epilepsy is considered a feasible approach for patients with drug-resistant epilepsy [2], [3], as researchers aim to effectively diagnose imminent or ongoing seizures via EEG, enabling timely preventive and protective measures to significantly reduce morbidity. Generally, the EEG signals of epilepsy patients can be categorized into four distinct phases: interictal (between seizures), preictal (immediately prior to a seizure), ictal (during a seizure), and postictal (immediately following a seizure). The different electrophysiological characteristics of these phases can be utilized to detect and predict seizure activity [4].

In recent years, advancements in artificial intelligence and wearable devices have led to significant progress in EEG-based epilepsy diagnostics. Many of these approaches involve training deep neural networks on pre-existing EEG data in cloud environments, with the models then deployed on edge devices for offline diagnosis [5]. However, epilepsy is a highly heterogeneous condition with varying types and underlying causes [6]. Due to challenges in acquiring physiological signals—such as the fact that available datasets often lack diversity and are collected under controlled conditions [7]—the neural networks trained on this data often fail to capture the full complexity of epilepsy, resulting in suboptimal detection or prediction performance on edge devices [8]. This problem is compounded as physiological conditions change over time, degrading model performance. In addition, continuously updating models by transferring large volumes of recollected data to the cloud is often impractical due to high cost and data privacy risks.

On-device learning presents an alternative paradigm, adapting the model directly on the user's hardware to bypass these data transmission issues. However, training deep neural

Received 15 December 2024; revised 26 August 2025; accepted 9 December 2025. Date of publication 15 December 2025; date of current version 6 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62404037 and Grant 623B2085, in part by Hebei Natural Science Foundation under Grant F2024501017, and in part by the Fundamental Research Funds for the Central Universities under Grant Z2024-043. (Shanjin Li and Di Wu contributed equally to this work.) (Corresponding author: Shiqi Zhao.)

Shanjin Li and Shiqi Zhao are with Northeastern University, Shenyang 110819, China, and also with the Hebei Key Laboratory of Marine Perception Network and Data Processing, Northeastern University, Qinhuangdao, Hebei 066004, China (e-mail: shanjinli@stumail.neu.edu.cn; zhaoshiqi@qhd.neu.edu.cn).

Di Wu, Jie Yang, and Mohamad Sawan are with the Center of Excellence in Biomedical Research on Advanced Integrated-on-chips Neurotechnologies (CenBRAIN Neurotech), School of Engineering, Westlake University, Hangzhou 310030, China, and also with the Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou 310024, China (e-mail: wudi@westlake.edu.cn; yangjie@westlake.edu.cn; sawan@westlake.edu.cn).

Digital Object Identifier 10.1109/JBHI.2025.3643602

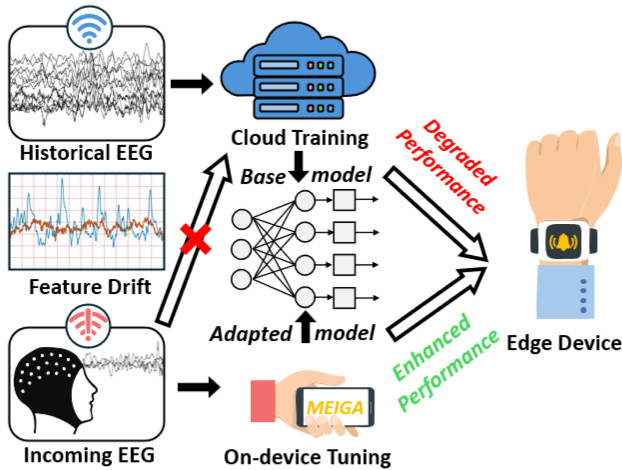


Fig. 1. System-level illustration of the proposed MEIGA framework. Conventional approaches train a base model on historical EEG data for deployment on edge devices. Performance degradation may occur due to feature drift between incoming and historical EEG data. MEIGA addresses this challenge by adapting the pre-trained model in an on-device manner, improving performance in real-world applications.

networks on edge devices requires substantial computational power and memory, which conflicts with the practical constraints of portable, resource-limited devices [9]. Achieving a balance between improving model performance through on-device training and maintaining device practicality in terms of portability, size, and power remains a significant challenge.

To address these challenges, we propose a hardware-efficient, on-device learning framework named **Memory Efficient Intrinsic Gating Adaptation (MEIGA)**, aiming at enhancing seizure diagnosis performance in resource-constrained edge devices, as illustrated in Fig. 1. Our approach targets scenarios in which a base model is pre-trained on data from previous EEG sessions and subsequently fine-tuned on user-side hardware (e.g., wearable or edge devices) during intermediate sessions, thereby mitigating performance degradation caused by session-to-session variability. This framework simulates real-world usage scenarios for EEG-based seizure detection and prediction, where data characteristics can shift over time. Although distribution shifts across sessions, the task and label space remain consistent [10], allowing the model to be reused and minimally adapted via a lightweight module for the target session, without full retraining.

Specifically, given a pre-trained model (which could be any arbitrary neural network architecture), we introduce lightweight adaptors with a small number of parameters atop the frozen pre-trained baseline. To update the model, we leverage Direct Feedback Alignment (DFA), training only the parameters of the adaptor network while keeping the baseline model's parameters fixed. This approach circumvents the backward locking constraint of traditional backpropagation, significantly reducing memory usage and computational overhead by enabling parallel transmission of loss values within the adaptor network. As a result, it offers a practical solution for on-device learning in portable devices, optimizing both performance and resource efficiency.

We rigorously assess the effectiveness of our proposed framework using the CHB-MIT epilepsy dataset [11] and the American Epilepsy Society Seizure Prediction Challenge (AES) Intracranial EEG (iEEG) dataset [12] for both seizure detection and prediction tasks. Extensive experiments show that our approach significantly outperforms traditional deployment paradigms, highlighting its advantages in real-world scenarios. Our key contributions are summarized as follows:

- 1) We propose **MEIGA**, a novel on-device learning approach that allows for real-time adaptation to session-to-session variations in EEG data, mitigating performance degradation over time for seizure diagnosis without the need for large-scale data recollection or cloud-based model updates, making it practical for resource-constrained edge devices.
- 2) To address the computational limitations of edge devices, we introduce a lightweight **MEIGA** adaptor network combined with DFA, significantly reducing memory usage and computational overhead compared to traditional backpropagation;
- 3) Extensive experiments validate the effectiveness of the proposed **MEIGA** framework, achieving state-of-the-art performance in seizure prediction and detection tasks while utilizing only a small fraction of the model parameters for updates.

II. RELATED WORKS

A. EEG-Based Seizure Stage Classification

Seizure episodes can severely impact patients, particularly affecting motor functions and consciousness [13]. Given these risks, accurate and timely detection of epilepsy stage transitions is crucial for patient safety, making it a central focus of extensive research. Over the years, researchers have developed various methodologies for data acquisition, pre-processing, feature extraction, and classification models, leading to significant advancements in the field [14].

Traditional approaches often rely on manual feature extraction, targeting features across the time domain [15], frequency domain [16], time-frequency domain [17], and nonlinear characteristics [18]. Models like support vector machine (SVM) [19], [20] and multi-layer perceptrons (MLP) [21] are commonly employed in such settings. While these methods can enhance classification accuracy, they require domain expertise and often lack generalizability across diverse EEG datasets.

In recent years, deep learning has revolutionized seizure prediction, showcasing exceptional performance. Convolutional Neural Networks (CNNs) have been particularly effective due to their ability to automatically extract features. For example, Wu et al. introduced C2SP-Net, a joint compression and classification network for EEG-based seizure prediction, achieving 92.5% accuracy and a low false prediction rate of 0.09/h on the CHB-MIT dataset [22].

Complementing CNN-based approaches, Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, have also been extensively used for their capability to model temporal dependencies in EEG signals.

Tsiouris et al., for instance, proposed a two-layer LSTM network trained on the CHB-MIT dataset, achieving a sensitivity of 99.86% and an FPR as low as 0.02 false alarms per hour with a 120-minute preictal window [23]. Hybrid models combining the strengths of CNNs and RNNs have gained traction. Daoud et al. proposed a patient-specific method using CNN and Bidirectional LSTM (Bi-LSTM) for long-term EEG data, achieving a sensitivity of 99.72% and a false alarm rate as low as 0.004 per hour [24].

Beyond CNNs and RNNs, attention-based architectures have emerged as strong contenders. Transformers [25], initially introduced for machine translation, have become state-of-the-art in many natural language processing tasks and are increasingly applied to seizure prediction. The core of Transformer architectures is the attention mechanism, specifically Scaled Dot-Product Attention, defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (1)$$

where Q , K and V denote queries, keys, and values, respectively, with $\sqrt{d_k}$ serving as a scaling factor, and T denoting the transpose operation [25]. For seizure prediction, Godoy et al. introduced Temporal Multi-Channel Vision Transformers (TMC-ViT), achieving state-of-the-art accuracy (95.73%), AUC (97.55%), and sensitivity (96.46%) using raw EEG data [26].

Three primary model training schemes are commonly employed for epilepsy detection and prediction: patient-specific, patient-independent, and patient-generic models. Patient-specific models have demonstrated superior clinical performance by tailoring predictions to individual patients. For example, Xu et al. [27] achieved 90.54% sensitivity, an AUC of 0.88, and a false prediction rate of 0.11/h on four CHB-MIT patients. Similarly, Zhang et al. [28] achieved sensitivity and specificity of 97.8% and 99.5%, respectively, with latency under 1 s using guided time-channel averaging (GTCA) on the CHB-MIT dataset. However, the limited availability of patient-specific data constrains these models. Patient-independent models involve pretraining on a pool of patients and fine-tuning on a left-out patient. This approach allows better calibration to individual characteristics and is widely applied in seizure detection and prediction tasks [29], [30]. Patient-generic models leverage pretraining on other patients to generalize performance across unseen patients, commonly used in seizure detection tasks [31], [32]. However, due to significant inter-patient variability in EEG biomarkers, achieving high accuracy remains challenging.

B. Adaptive Learning for Seizure Classification on Edge Devices

The advancements in seizure stage classification highlight the growing potential of AI methods for real-world applications. However, deploying AI models on edge devices for long-term seizure monitoring presents significant challenges. In practical scenarios, models cannot access the entirety of a user's historical data during training, limiting their ability to generalize to future, unseen conditions. Moreover, post-deployment, models face difficulties in incorporating new data, often requiring complete

retraining or leaving the new data unprocessed, which undermines adaptability and performance [33].

These challenges are exacerbated by the dynamic and non-stationary nature of EEG signals, where shifts in data distributions over time can lead to performance degradation. Such issues may result in false alarms or missed detections, causing unnecessary panic or harm to patients and rendering these models unreliable in real-world use. Addressing this problem requires models capable of adapting to new data streams without the need for extensive retraining. Online learning and incremental learning have emerged as promising solutions to tackle these challenges.

Online learning enables models to process and learn from individual data instances sequentially, unlike traditional batch-based machine learning approaches [34]. This paradigm has been extensively researched in signal domains such as EMG, ECG, and EEG [35], [36], [37], [38], [39], with successful real-world applications. For instance, the Stochastic Online Unsupervised Learning (SOUL) classifier [37] demonstrated sensitivities of 97.5% and 97.9% on the CHB-MIT scalp EEG dataset and iEEG dataset, respectively. The approach employed a stochastic gradient descent-based algorithm implemented on custom hardware, achieving energy efficiency of 1.5 nJ/classification. Similarly, Benatti et al. [38] developed a wearable electromyographic gesture recognition system achieving 85% accuracy on 11 gestures. Using a hardware-efficient algorithm on the PULP platform, the system consumed 10.04 mJ for training and 83.2 μ J per classification, demonstrating a scalable, fully embedded online learning framework. Another example is the adaptive EMG decomposition method proposed by Guerra et al. [39], which utilized Kullback–Leibler divergence and kurtosis as metrics. This approach achieved real-time adaptation, with sensitivity and precision rates exceeding 90% in over 80% of cases, validated across simulated and experimental data.

Incremental learning, also known as continual or lifelong learning, addresses the issue of catastrophic forgetting, where models trained on new data lose the ability to retain knowledge from earlier training stages. Incremental learning has shown great promise in fields like computer vision [40] and medical imaging [41]. In seizure prediction, Zhang et al. [42] introduced an incremental logistic regression model with sparse group constraints, achieving an average AUC of 0.686 across multiple subjects. Deng et al. [43] proposed the ER-CM method, which mitigates catastrophic forgetting using centroid matching for domain incremental learning, significantly enhancing inter-patient performance. Wang et al. [44] developed a channel-increment strategy using 1D-CNN on iEEG data, achieving 90.18% sensitivity and 94.42% accuracy on the Freiburg iEEG database by incrementally feeding data with increasing channels.

While these methods have made substantial contributions, several limitations persist in their application to seizure prediction and classification on edge devices. Many solutions [35], [38], [39], [43] focus on broader healthcare challenges or catastrophic forgetting without addressing the computational constraints of edge devices. Furthermore, complex algorithms like those proposed by Kiyasseh et al. [40] are impractical for resource-constrained environments. To address these gaps, our

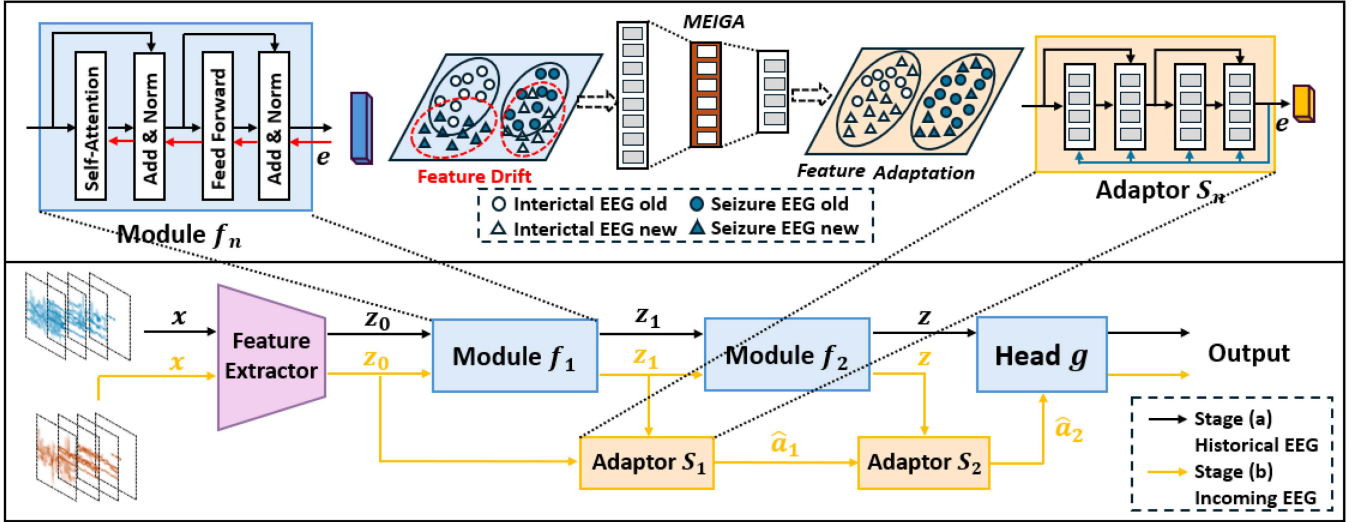


Fig. 2. Overview of the proposed MEIGA framework. The MEIGA framework comprises two stages: (a) Base Model Pre-training Stage: A feature extractor and additional modules of arbitrary design are trained on historical EEG data to create a baseline model, which is deployed on edge devices for seizure diagnostics. However, performance degrades over time due to shifts in patient biomarkers; (b) MEIGA Stage: To address performance degradation within the constraints of edge devices, lightweight adaptors are employed to adapt features in a low-dimensional intrinsic space using incoming EEG data, while keeping the feature extractor and modules frozen. This adaptation ensures improved diagnostic performance. Both modules and adaptors are architecture-agnostic; in this study, we use an attention module and an MLP-based adaptor as a proof of concept.

MEIGA approach is tailored to handle the temporal variability of EEG data, computational constraints, and adaptability requirements for edge-device deployment. The framework leverages state-of-the-art advancements in online and incremental learning to achieve robust, resource-efficient seizure classification and prediction.

III. METHODS

Addressing the performance degradation of deployed models due to physiological variations over time is a critical challenge in diagnosing neurodegenerative diseases based on physiological signals, such as epileptic seizures. To tackle this issue and enhance the practicality of seizure diagnostics, we propose a novel framework, as shown in Fig. 2, which includes a pre-training stage followed by an on-device rapid tuning stage, named Memory Efficient Intrinsic Gating Adaptation. During the pre-training phase, we train a neural network with any arbitrary design using previously collected seizure session data for the target subject. In the on-device tuning stage, we keep the pre-trained backbone frozen while performing training a lightweight adaptor directly on-device using intermediate sessions collected post-deployment. This approach is designed to be hardware-efficient and practical for edge deployment.

A. Base Model Pre-Training

Let $\mathcal{X} \in \mathbb{R}^{N \times C}$ be the multi-channel input EEG signal of length N with C channels, we then define a transformer based neural network f as the backbone model for feature extraction. The intermediate neural representation output by each transformer block of the network is denoted as $z_i = f_i(z_{i-1})$, where i indicates the corresponding layer or module number and z represents the last output right before the prediction head g

for seizure analysis. During the pre-training stage, we train the network in a supervised manner with EEG data collected from subjects and annotated by medical experts. Let $y \in \mathcal{Y}$ be the corresponding label of x either for seizure detection or prediction and the model aims to decode the corresponding neural state y from the input signals x . We utilize the cross entropy loss as the objective function during training:

$$\mathcal{L} = - \sum_i y^i \cdot \log(\text{softmax}(g(z^i))), \quad (2)$$

where the softmax operation maps the logits from the prediction head g to a probabilistic distribution.

B. Memory Efficient Intrinsic Gating Adaptation

With the deployment of the pre-trained model on end devices, we propose a MEIGA stage that adapts the model on-device with upcoming EEG measurements related to seizure analysis for an enhanced performance. Our intuition is that although data variation exists among sessions, we can adapt the extracted neural feature in the intrinsic dimension (much smaller than the original feature dimension) to strike a balance between performance and computation constraints. Specifically, we keep the pre-trained model weights frozen and introduce another light-weight MEIGA adaptor to modify the intermediate representations z_i , demonstrated in Algorithm 1.

The adaptation is performed by first transforming the intermediate features to an intrinsic space $z_i \in \mathbb{R}^{N_i \times C_i} \Rightarrow \hat{z}_i \in \mathbb{R}^{N_i \times \hat{C}_i}$. For simplicity, the intrinsic dimension is assumed to be acquired with a compression factor r , that is $\hat{C}_i = \frac{1}{r} \cdot C_i$. A simple linear layer or an MLP could be utilized for downsampling. For instance, given an intermediate feature of z_i with a representation dimension of 256, a compression factor of $r = 4$

Algorithm 1: MEIGA adaptor.

```

1: Initialize:  $\hat{a}_0 = \hat{z}_0 = \text{compress}_0(z_0)$ 
2: for  $i = 1$  to  $M - 1$  do
3:    $\hat{z}_i = \text{compress}_i(z_i)$ 
4:    $a_i = (1 - \gamma_i) \cdot \hat{z}_i + \gamma_i \cdot \hat{a}_{i-1}$ 
5:    $\hat{a}_i = S_i(a_i)$ 
6: end for
7:  $\hat{z} = \text{compress}_M(z)$ 
8:  $a_M = (1 - \gamma_M) \cdot \hat{z} + \gamma_M \cdot \hat{a}_{M-1}$ 
9:  $\hat{a}_M = \text{expand}(a_M)$ 
10: return  $\hat{a}_M$ 

```

will be mapped to an intrinsic dimension of 64. We then apply a gating operation to integrate the mapped intermediate features output by the backbone model:

$$a_i = \begin{cases} (1 - \gamma_i) \cdot \hat{z}_{i+1} + \gamma_i \cdot \hat{z}_i, & \text{if } i = 0, \\ (1 - \gamma_i) \cdot \hat{z}_{i+1} + \gamma_i \cdot \hat{a}_{i-1}, & \text{if } i > 0. \end{cases} \quad (3)$$

where $\gamma_i = \text{sigmoid}(\frac{\alpha_i}{T})$ denotes a learnable weighting gating coefficient [45]. On top of the gated representation a_i , we further utilize a lightweight adaptation module S_i to transform a_i such that it effectively accommodates the biomarker variation of future epileptic seizures. The design of the adaptation module can be tailored to match the backbone architecture, such as encoder modules for Vision Transformers or linear modules for MLPs. Our proposed MEIGA framework adapts the intermediate features output by the backbone model iteratively module by module, with the last adapted intrinsic representation a_M projected back to the original feature space $a_i \in \mathbb{R}^{a_i \times \hat{C}_M} \Rightarrow \hat{a}_M \in \mathbb{R}^{N_M \times C_M}$, such that the dimension of the final adapted representation aligns with the prediction head trained in the pre-training stage seamlessly. We again utilize the cross entropy as the training objective described in Section III-A.

C. Efficient Training of MEIGA

The backpropagation (BP) algorithm is widely recognized as one of the most effective methods for training deep neural networks. However, its inherent backward locking mechanism mandates the sequential processing of layers, thereby limiting opportunities for parallelization [46]. Moreover, BP necessitates storing the input and output values for all layers during both the forward and backward passes, leading to a substantial increase in memory consumption. DFA has emerged as a promising alternative to address these limitations, as demonstrated in Fig. 3. Its tolerance to hardware imperfections further enhances its appeal, making it particularly suitable for hardware/software co-design environments where efficiency and robustness are critical [47]. Assuming that the backbone module contains a MLP subnetwork integrated with our proposed MEIGA adaptor, the final output is defined as $y_{\text{pred}} = \sigma(W_{\text{head}} \cdot \hat{a}_M + b)$, where σ represents the activation function, W_{head} is the weight matrix of the prediction head, and b denotes the bias term. The gradients of the loss with respect to various parameters in the network can be expressed as follows:

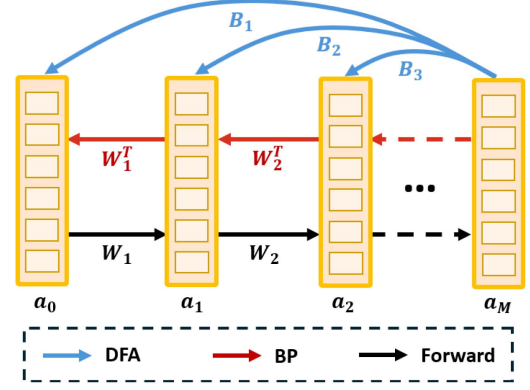


Fig. 3. Comparison of BP and DFA algorithms. W_i represents the weights updated during adaptation, while B_i denotes randomly generated projection matrices used for error mapping. Unlike BP, which relies on the traditional chain rule for gradient computation, DFA bypasses this process by directly computing gradients through random projections from the final output layer, enabling more efficient parallel updates.

1. Gradient with respect to W_{head} :

$$\frac{\partial L}{\partial W_{\text{head}}} = \left(\frac{\partial L}{\partial y_{\text{pred}}} \odot \sigma' \right) \hat{a}_M^\top$$

2. Gradient with respect to W_{expand} :

$$\frac{\partial L}{\partial W_{\text{expand}}} = \left(W_{\text{head}}^\top \left(\frac{\partial L}{\partial y_{\text{pred}}} \odot \sigma' \right) \right) a_M^\top$$

3. Gradient with respect to γ_M :

$$\frac{\partial L}{\partial \gamma_M} = \left(\left(\frac{\partial L}{\partial y_{\text{pred}}} \odot \sigma' \right) W_{\text{head}} W_{\text{expand}} \right) (-\hat{z} + \hat{a}_{M-1})$$

4. Gradient with respect to W_{compress_M} :

$$\frac{\partial L}{\partial W_{\text{compress}_M}} = \left(\left(\frac{\partial L}{\partial y_{\text{pred}}} \odot \sigma' \right) W_{\text{head}} W_{\text{expand}} \gamma_M \right) a_{M-1}^\top$$

where W_{expand} and W_{compress_M} represent the weights associated with the $\text{expand}(\cdot)$ and $\text{compress}_M(\cdot)$, respectively, as defined in the algorithm.

The conventional backpropagation method involves lengthy chain rule calculations, resulting in high memory usage due to the storage of intermediate activations and gradients. To address this, DFA provides a more memory-efficient alternative by aligning the computed gradients with a random projection matrix, B . This significantly simplifies the gradient propagation and mitigates the computational burden.

Under DFA, the gradients are modified as follows:

1. Gradient with respect to W_{head} :

$$\frac{\partial L}{\partial W_{\text{head}}} = \left(\frac{\partial L}{\partial y_{\text{pred}}} \odot \sigma' \right) \hat{a}_M^\top$$

2. Gradient with respect to W_{expand} :

$$\frac{\partial L}{\partial W_{\text{expand}}} = \left(\left(B \frac{\partial L}{\partial y_{\text{pred}}} \right) \odot \sigma' \right) a_M^\top$$

3. Gradient with respect to γ_M :

$$\frac{\partial L}{\partial \gamma_M} = \left(\left(B \frac{\partial L}{\partial y_{\text{pred}}} \right) \odot \sigma' \right)^\top (-\hat{z} + \hat{a}_{M-1})$$

4. Gradient with respect to W_{compress_M} :

$$\frac{\partial L}{\partial W_{\text{compress}_M}} = \left(\left(B \frac{\partial L}{\partial y_{\text{pred}}} \right) \odot \sigma' \right) a_{M-1}^\top$$

In these formulations, B represents a fixed random projection matrix, with its size determined by the length of the error vector and the output dimension of each layer. This matrix facilitates the direct mapping of errors from the output layer to individual layers, thereby bypassing the sequential gradient pathway required in traditional backpropagation. In practice, [48] suggested that layers of identical size can share the same projection matrix, and the entire backward pass of DFA can be implemented by initializing a single random matrix in the range $[-1, 1]$ for the largest layer with the class size of the classification task, while smaller layers use fixed slices of this matrix. This strategy not only reduces memory requirements but also ensures consistency in the random projection mechanism across the network.

The stability of this random projection mechanism has also been validated across different architectures. While earlier works noted that CNNs are difficult to train directly with DFA, practical workarounds involved pre-training convolutional layers and applying DFA to the fully connected layers [49], [50]. More recently, [51] showed that DFA can train advanced architectures such as graph networks and Transformers. Although a performance gap with BP remains, these results suggest that random projection is sufficiently stable across complex tasks, and further progress may depend on new training strategies.

Building on this foundation, our method MEIGA eliminates the layer-by-layer dependency, significantly reducing memory usage and computational complexity, thereby enhancing its compatibility with hardware constraints.

The application of DFA, as shown in Fig. 4, significantly improves memory efficiency during both the forward and backward passes of training. In the forward pass, memory usage is reduced by limiting tuning to the adaptor network, eliminating the need to store the most hidden outputs within the backbone model. In the backward pass, DFA bypasses the traditional chain rule by directly computing gradients through random projections from the final output layer. This streamlined approach not only conserves memory but also enhances the overall efficiency of the training process, making it particularly advantageous for deep neural networks.

IV. EXPERIMENTS

For this study, we utilized the CHB-MIT epilepsy dataset and the AES dataset.

- **CHB-MIT dataset:** This is a widely recognized resource in epilepsy research, containing EEG recordings from pediatric patients at Boston Children's Hospital. The patients were monitored over several days without antiseizure medications to assess their suitability for neurosurgery. The dataset includes 23 cases from 22 patients, aged between

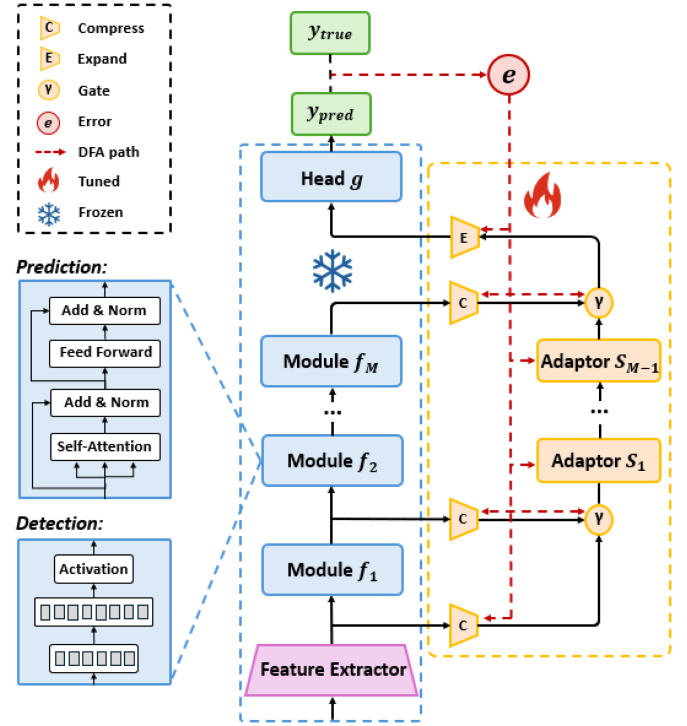


Fig. 4. Demonstration of Efficient Training of MEIGA Adaptors Using DFA. This figure illustrates the adaptation process of MEIGA adaptors, where the training loss is minimized through DFA. The compress/expand operations represent transformations to and from the intrinsic (lower-dimensional) space, while γ acts as a gating mechanism. During adaptation, the backbone model remains frozen, and only the adaptor parameters are updated. The training error e is directly propagated to all learnable components within the adaptor, bypassing the traditional backpropagation chain rule for efficient on-device learning.

1.5 and 22 years. EEG signals were recorded at a sampling rate of 256 Hz using scalp electrodes.

- **AES dataset:** This consists of intracranial EEG (iEEG) recordings collected from five canines and two human patients. The canine recordings were sampled at 400 Hz, with four dogs monitored using 16 electrodes and one dog (Dog 5) using 15 electrodes. The human recordings were acquired at a higher sampling rate of 5000 Hz, using 15 electrodes for one patient and 24 electrodes for the other. This dataset provides a diverse set of intracranial recordings with distinct sampling configurations across species, supporting robust evaluation of seizure prediction models.

To ensure consistency in our analysis, we focused on a standardized set of electrodes in the CHB-MIT dataset: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, P7-T7, T7-FT9, FT9-FT10, and FT10-T8. For the AES dataset, the two human patients were excluded due to the significantly different signal acquisition setup.

In both prediction and detection tasks, we focused on distinguishing different seizure states by defining a “leading seizure” as the first clinical seizure preceded by at least four hours of interictal activity [52]. To the best of our knowledge, there has

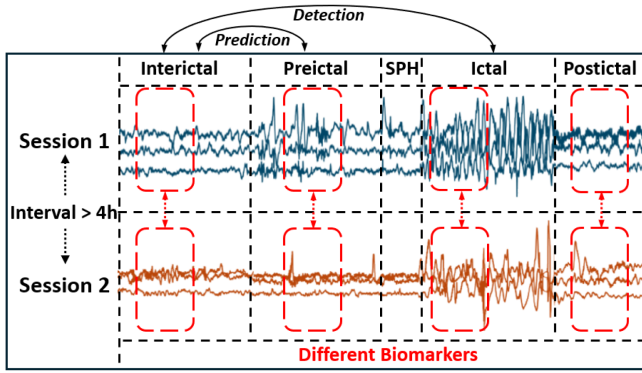


Fig. 5. Illustration of the seizure prediction and detection tasks along-side biomarker variation across sessions. The seizure prediction task involves distinguishing the interictal state from the preictal state, while the seizure detection task classifies the ictal state against all other states. The example highlights biomarker variation between sessions, demonstrating that even the same states exhibit noticeable differences across sessions.

TABLE I
SEIZURE FREE-TIME INTERVALS FOR EACH SUBJECT

Subject	Lead seizure index	Seizure-free period
01	1, 3	11.0h
04	1, 2	8.4h
05	1, 2	7.1h
07	1, 3	23.3h
08	3, 5	7.8h
09	1, 2	5.4h
10	1, 2	16.1h

been limited research on using sessions as a basis for splitting training and testing data in seizure classification. Even within the same patient, biomarkers can vary significantly over time—for instance, across different times of the day—making pretrained models ineffective to temporal variations.

To contend with this challenge, we utilized the first leading seizure as the initial session for pretraining and the second leading seizure for fine-tuning, as illustrated in Fig. 5, ensuring the model could effectively adapt to temporal variations. For the CHB-MIT dataset, we selected seven patients (01, 04, 05, 07, 08, 09, and 10) who had at least two sessions with leading seizures. The seizure index in Table I corresponds to the numbering of the two leading seizures in each patient’s dataset, with the time intervals between them ranging from 5 to 23 hours. This setup provided a robust foundation for pretraining and fine-tuning the model to account for temporal variability in the data. For the AES dataset, the recordings were already organized around leading seizures as part of the challenge design. The raw EEG signals were used directly without additional feature extraction, relying on the model’s capacity for automatic feature learning. This approach leverages the model’s proven effectiveness in handling the inherent variability and noise in EEG signals, as demonstrated in prior research [53].

To evaluate the model’s performance on the CHB-MIT dataset, we employed several commonly used metrics: sensitivity (SS), specificity (SP), accuracy (ACC), and F1-score.

For the AES dataset, following the original challenge protocol and considering its clinical relevance, we report SS and the area under the ROC curve (ROC AUC), capturing both seizure detectability and overall discriminative performance.

A. Seizure Prediction

1) *Data Setting*: Seizure Prediction Horizon (SPH) and Preictal Interval Length (PIL) are critical factors in seizure prediction [54]. For the CHB-MIT dataset, we set the Seizure Prediction Horizon (SPH) to 5 minutes and the Preictal Interval Length (PIL) to 30 minutes, following settings adopted in prior works [22], [26], [55]. These values serve as benchmark parameters that determine which segments of the recordings are selected for experiments and were chosen to enable fair comparison with other studies. Additionally, the interictal state was extracted from the one-hour period preceding the SPH, allowing for a thorough analysis of the transition from non-seizure to seizure-prone conditions.

In terms of the AES dataset, the PIL (1 h) and SPH (5 minutes) were predefined by the challenge. Preictal data are provided as six consecutive files per seizure, which we concatenate and treat as a continuous window. Interictal data are supplied as discontinuous 10-minute clips sampled at random times, preventing contiguous extraction. Accordingly, we randomly sampled interictal clips totaling twice the preictal duration for each seizure to construct our dataset.

The data was initially split into an 8:2 ratio for training and test sets, respectively. To segment the EEG data into standardized intervals, a window size of 20 seconds was employed. For the CHB-MIT training set, a 95% overlap was applied to the preictal state and a 90% overlap to the interictal state. This overlap strategy ensured a balanced representation of both states, which is crucial for effective seizure prediction, particularly since the interictal period is twice the duration of the preictal period. While for the AES dataset, we applied class-balanced weighting during training to address the imbalance between preictal and interictal samples.

2) *Architecture*: One of the primary advantages of Transformers is their capacity to exploit parallel computing, which significantly accelerates training times. Leveraging this strength, the Vision Transformer (ViT) has achieved remarkable success in computer vision tasks, such as image classification and processing [56]. In this study, we adapted the ViT architecture for the specific challenge of seizure prediction, as depicted in Fig. 6. For the embedding stage, we utilized a 1D convolutional neural network (1D-CNN) along the time axis to segment the input EEG signals into patches, employing a kernel size and stride of 8 to ensure non-overlapping segments. A classification token was appended to aggregate global information. To address the positional information loss inherent in segmentation, we incorporated learnable positional embeddings, which were combined with the patch embeddings to encode spatial context.

The resulting sequence, enriched with both patch and positional embeddings, was fed into the Transformer encoder, which consisted of four stacked blocks. Each block included multi-head self-attention mechanisms with a head dimension of 32 and four

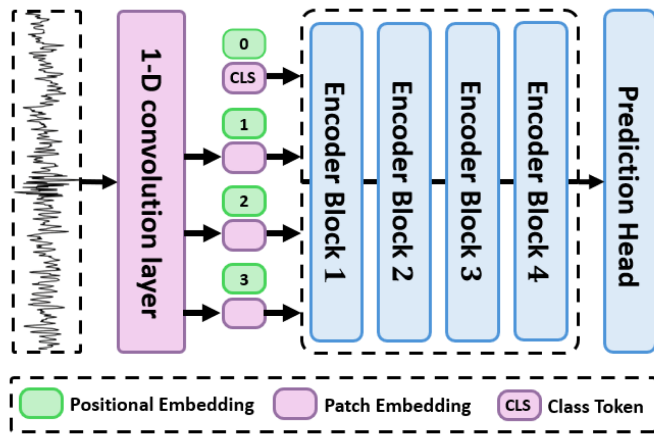


Fig. 6. Backbone architecture for the seizure prediction task. A 1-D convolutional layer is employed as the patch embedding layer, where the output tokens are augmented with positional embeddings and concatenated with a classification token. The resulting representations are processed through encoder blocks, as defined as modules in Fig. 2, before being passed to the prediction head for final seizure prediction.

attention heads, feed-forward layers with a hidden dimension of 128, residual connections, and layer normalization. Together, these blocks facilitated efficient and hierarchical feature extraction. Finally, the output from the Transformer encoder was processed through a linear layer to generate the final predictions.

In the AES experiments, the backbone employed the same architecture, with the only modification being an increased head dimension of 128 and a hidden dimension of 512 to accommodate the higher-resolution intracranial recordings. This adjustment also allowed us to investigate whether the proposed framework can scale effectively to a larger network while maintaining its predictive capability.

For integrating the MEIGA adaptor, we selected the encoder module due to its proven ability to extract meaningful features, especially in distinguishing between the highly similar preictal and interictal states. Furthermore, this choice ensured structural consistency with the backbone model. To enhance efficiency, we applied a compression factor of 8 to the hidden state dimensions, balancing computational demands with model performance.

3) Implementation Details for Prediction: For pre-training, we used a Stochastic Gradient Descent (SGD) optimizer with a learning rate of 3×10^{-3} , a batch size of 32, and a dropout rate of 0.1. During the tuning stage, the model was fine-tuned using the Adam optimizer, targeting only the parameters within the MEIGA adaptor with a learning rate of 3×10^{-3} , betas = (0.9, 0.999), and eps = 1×10^{-9} . The training was conducted over 50 epochs with a dropout rate of 0.3, using a Cosine Annealing Learning Rate scheduler with T_{max} set to 50 epochs.

In the AES experiments, the same configuration was applied, except that both pre-training and tuning used a lower learning rate of 5×10^{-4} , and training loss was monitored to enable early stopping.

4) Result Comparison: Tables II and III summarize the experimental results for the seizure prediction tasks on the AES and CHB-MIT datasets, respectively, evaluating four training schemes: a pretrained model trained on historical EEG data,

TABLE II
COMPARISON OF RESULTS ACROSS DIFFERENT TRAINING SCHEMES ON THE CHB-MIT DATASET

Subject	Pretrained		Direct Trained		MEIGA (BP)		MEIGA (DFA)			
	ACC	SS	ACC	SS	ACC	SS	ACC	SS	SP	F1
01	57.41%	0.8333	98.15%	1.0000	94.44%	0.9444	92.59%	0.9444	0.9167	0.895
04	33.33%	1.0000	90.74%	0.8333	92.59%	0.9444	81.48%	0.7778	0.8333	0.737
05	37.04%	0.7778	85.19%	0.8611	81.48%	0.8333	85.19%	0.8889	0.8333	0.800
07	66.67%	0.3889	85.19%	0.6667	87.04%	0.7222	87.04%	0.9167	0.7778	0.791
08	40.74%	0.0000	90.74%	0.8889	83.33%	0.6667	81.48%	0.7778	0.8889	0.778
09	53.70%	0.0556	83.33%	0.9167	81.48%	0.8611	87.04%	0.8333	0.8889	0.811
10	46.30%	0.3889	100.00%	1.0000	96.30%	0.9444	92.59%	0.8333	0.9722	0.882
Average	47.88%	0.4921	90.48%	0.8810	88.09%	0.8452	86.77%	0.8532	0.8730	0.813

TABLE III
COMPARISON OF RESULTS ACROSS DIFFERENT TRAINING SCHEMES ON THE AES DATASET

Subject	Pretrained		Direct Trained		MEIGA (BP)		MEIGA (DFA)	
	SS	AUC	SS	AUC	SS	AUC	SS	AUC
Dog1	0.200	0.549	0.943	0.942	0.971	0.857	0.857	0.796
Dog2	0.000	0.967	0.971	0.993	0.971	1.000	0.943	0.985
Dog3	0.971	0.999	1.000	1.000	1.000	0.999	1.000	0.998
Dog4	0.143	0.973	1.000	1.000	1.000	1.000	0.886	1.000
Dog5	0.829	0.938	0.914	0.988	0.914	0.961	0.943	0.982
Average	0.429	0.885	0.966	0.985	0.971	0.963	0.926	0.952

a model trained directly on new EEG data (performance upper bound), and the MEIGA adaptor with two optimization algorithms—BP and DFA. Specifically, the pretrained model, trained exclusively on the first session, simulates the inference scenario on edge devices without any further on-device retraining. In contrast, the directly trained model is optimized on the second session using backpropagation, representing the performance upper bound. The MEIGA adaptor builds on the pretrained model as its backbone and employs either BP or DFA for fine-tuning, emphasizing optimization for edge-device compatibility. This design ensures a consistent comparison across all schemes while showcasing the trade-offs in performance and applicability for each approach.

As shown in Table II, the pretrained model trained on historical EEG data achieved the lowest average accuracy of 47.88% and a sensitivity of 0.4921. This underperformance can be attributed to the substantial biomarker variability over time between sessions. Notably, for subjects 04 and 08, the model exhibited extreme behavior: all segments from subject 04 were misclassified as preictal, while the majority of segments from subject 08 were identified as interictal. These results underscore the challenges of generalizing from one session to another due to temporal and physiological variability.

In contrast, the directly trained model achieved the highest average accuracy of 90.48% and a sensitivity of 0.881, demonstrating the advantages of session-specific training. However, this approach required a substantial parameter count (77,314 parameters), making it less suitable for resource-constrained edge devices. Furthermore, training from scratch for each session is impractical for real-world applications, emphasizing the need for efficient adaptation strategies. By introducing DFA training on the MEIGA adaptor, which contains only 3,908 parameters—approximately 5.05% of the backbone model—significant performance improvements were achieved over the pretrained model. The adaptor trained with DFA yielded an average accuracy of 86.77% and a sensitivity of 0.8532, demonstrating its effectiveness in addressing temporal variability between sessions. To further evaluate the MEIGA adaptor's performance,

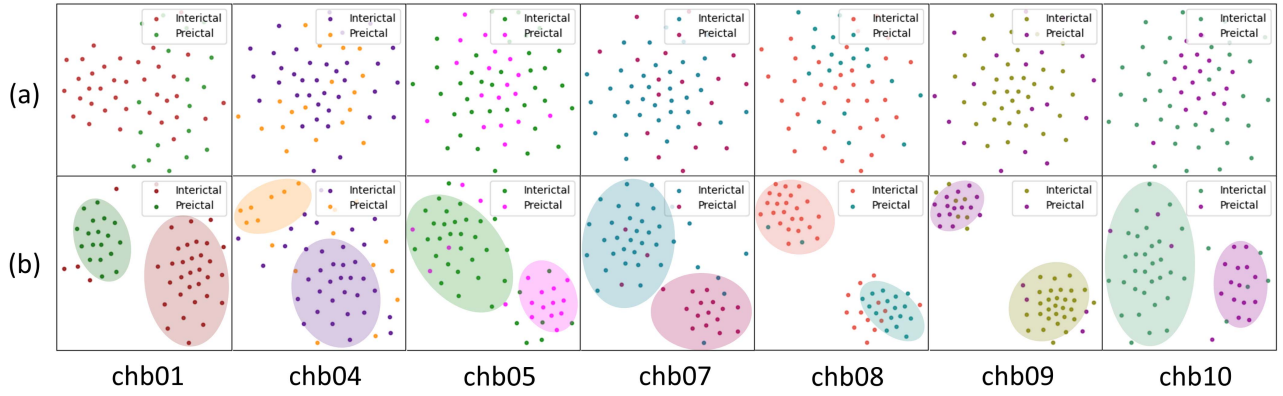


Fig. 7. Visualization Comparison of Representations Using t-SNE for Seizure Prediction Task: (a) The pre-trained model's representation on incoming EEG data (Session 2) exhibits poor linear separation, primarily due to variations in patient biomarkers between sessions; (b) In contrast, the MEIGA-adapted model demonstrates a clear clustering effect, effectively distinguishing between interictal and preictal states, highlighting the effectiveness of MEIGA in adapting to session-to-session variability.

an ablation study was conducted using BP for training—an approach unsuitable for edge devices but viable on standard training systems. The BP-trained MEIGA adaptor achieved an average accuracy of 88.09%, ranking second only to the directly trained model, and an average sensitivity of 0.8452, slightly higher than the DFA-trained adaptor.

These findings highlight the critical engineering trade-off central to our framework. The modest accuracy gap is an expected outcome of the underlying algorithms: BP calculates the exact gradient, establishing a performance ceiling but at a high computational and memory cost that is prohibitive for edge devices. In contrast, DFA approximates the gradient using random projections, sacrificing a small amount of accuracy for the significant efficiency required for practical on-device learning. This trade-off is what makes the MEIGA (DFA) approach a viable solution, not just a theoretical one. Importantly, MEIGA exhibits robustness and efficiency in identifying preictal patterns, underscoring its suitability for resource-constrained edge training environments.

Table III presents the performance of different setups on the AES dataset. The pretrained model again performed poorly, with an average sensitivity of 0.429 and an AUC of 0.885, highlighting the difficulty of transferring knowledge across sessions—particularly evident for Dogs 1, 2, and 4. In contrast, the directly trained model achieved the best AUC, establishing a stronger upper bound with consistently high performance across all subjects. The MEIGA adaptor, with 200,992 parameters (9.38% of the backbone size), provided an efficient alternative. When trained with BP, it achieved a sensitivity of 0.971 and an AUC of 0.963, while DFA yielded a sensitivity of 0.926 and an AUC of 0.952. These findings suggest that for larger networks, the MEIGA framework can mitigate model degradation by demonstrating strong adaptability to new sessions, while remaining competitive with direct training through substantial parameter reduction and preserved predictive performance.

5) Representation Visualization: In addition to numerical evaluation on the CHB-MIT dataset, we provide visualization results using t-Distributed Stochastic Neighbor Embedding (t-SNE) to analyze feature representations, as shown in

TABLE IV
COMPARISON OF INTER-CLASS DISTANCE AND SILHOUETTE SCORE BETWEEN PRE-TRAINED AND MEIGA-ADAPTED MODELS ON THE CHB-MIT DATASET

Subject	Pre-trained Model		MEIGA-adapted Model	
	Inter Dist.	Silhouette Score	Inter Dist.	Silhouette Score
01	116.10	0.1812 $\pm$ 0.0293	207269.89	0.6037 $\pm$ 0.0277
04	53.99	0.0094 $\pm$ 0.0326	114826.47	0.1608 $\pm$ 0.0202
05	53.09	−0.0214 $\pm$ 0.0126	93238.99	0.1968 $\pm$ 0.0338
07	77.82	0.0891 $\pm$ 0.0300	98063.90	0.2960 $\pm$ 0.0920
08	58.60	0.0061 $\pm$ 0.0335	104459.63	0.1871 $\pm$ 0.0545
09	61.10	0.0700 $\pm$ 0.0119	328305.81	0.3260 $\pm$ 0.0102
10	62.79	−0.0225 $\pm$ 0.0112	228875.05	0.4209 $\pm$ 0.0380
Average	69.07	0.0446 $\pm$ 0.0230	167862.82	0.3130 $\pm$ 0.0395

Fig. 7. Specifically, the features from the layer preceding the final prediction head of the models were projected into a two-dimensional space. Fig. 7(a) illustrates the feature representation of new EEG data produced by the pre-trained model. The separation between interictal and preictal states is poorly defined, with substantial overlap between the two classes, indicating that the pre-trained model struggles to effectively distinguish these states without further adaptation. In contrast, Fig. 7(b) depicts the representation generated by the MEIGA-adapted model. A clearer separation between interictal and preictal states is evident across the majority of test points, demonstrating the MEIGA framework's ability to enhance feature learning and classification performance. This visual improvement underscores the MEIGA framework's effectiveness in adapting the model to better distinguish between interictal and preictal states, ultimately contributing to its superior performance in seizure prediction tasks. We further quantified feature separability using inter-class distance and silhouette score. Inter-class distances were computed in the original feature space as the Euclidean distance between interictal and preictal centroids. Silhouette scores were computed on two-dimensional t-SNE embeddings, averaged over 10 random seeds. As shown in Table IV, the MEIGA-adapted model consistently achieved much larger inter-class distances than the pre-trained model, indicating improved separability. For example, in Subject 01, the inter-class distance rose from 116.10 to 207,269.89, and the silhouette score from

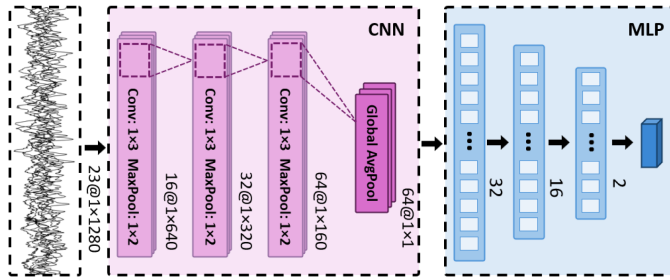


Fig. 8. Backbone network for the seizure detection task. The backbone architecture consists of three convolutional blocks, each followed by max-pooling layers. The convolution operations are confined to the time dimension, ensuring channel-wise separation of EEG signals. An average pooling layer, two fully connected layers, and a final linear layer with a softmax activation generate the output predictions.

0.1812 ± 0.0293 to 0.6037 ± 0.0277 . Similar large gains were observed in Subjects 07, 09 and 10, confirming that the adapted model yields more discriminative features and more compact, well-separated clusters.

B. Seizure Detection

1) Data Setting: To address the significant class imbalance in seizure detection, we downsampled the non-ictal stage to 10 minutes for each session. The dataset was also split into 80% for training and 20% for testing. EEG data were segmented using a window size of 5 seconds to create consistent and manageable samples. Within the training set, a 90% overlap was applied to the ictal state, while a 50% overlap was used for the non-ictal state, ensuring a balanced representation of seizure and non-seizure conditions.

2) Architecture: For the epilepsy detection task of distinguishing ictal states from non-ictal states, we utilized a straight-forward hybrid backbone model that combines CNNs with an MLP, as illustrated in Fig. 8. The CNN component consists of three convolutional layers, each with a kernel size of 3 and a stride of 1, designed to extract temporal features along the time channel. The output channels for the three convolutional layers are 16, 32, and 64, respectively. Each convolutional layer is followed by max-pooling with a kernel size and stride of 2, effectively reducing dimensionality while preserving critical features. A global average pooling layer is applied after the convolutional layers to aggregate the extracted features, reducing the spatial dimensions and preparing the data for subsequent processing. The MLP component includes three fully connected layers, which further refine the extracted features. These layers have an input-output configuration of 64, 32, and 16 dimensions, respectively. The first two layers incorporate ReLU activation functions and residual connections, enhancing the model's capacity for learning complex patterns. The final layer applies a softmax activation function to produce class probabilities for the prediction task.

During the tuning stage, the MEIGA adaptor is integrated into the MLP component to address limitations associated with DFA, particularly its tendency to degrade accuracy in CNN architectures [50]. The MEIGA adaptor is implemented as a

TABLE V
RESULTS COMPARISON BETWEEN DIFFERENT TRAINING SCHEMES FOR SEIZURE DETECTION

Subject	Pretrained	Direct Trained	MEIGA(BP)	MEIGA(DFA)
01	71.43%	100.00%	100.00%	100.00%
04	89.74%	97.44%	79.49%	92.31%
05	89.74%	100.00%	100.00%	100.00%
07	88.89%	100.00%	97.22%	97.22%
08	69.44%	97.22%	88.89%	91.67%
09	96.88%	100.00%	100.00%	100.00%
10	89.29%	92.86%	89.29%	92.86%
Average	85.06%	98.22%	93.56%	96.29%

series of linear transformations with activation functions, positioned after each linear layer in the MLP. This design mitigates DFA's challenges in training deeper networks, ensuring more robust learning. To maintain efficiency, the MEIGA adaptor employs a predefined compression factor that reduces the hidden dimensions of the backbone layers. This ensures the gating mechanism operates on features of a consistent reduced dimensionality, balancing computational efficiency with performance while enhancing the overall adaptability of the model.

3) Implementation Details: For pre-training, we utilized a Stochastic Gradient Descent (SGD) optimizer with a batch size of 64, a learning rate of 2×10^{-2} , and cross-entropy loss. During the tuning stage, the Adam optimizer was employed with a batch size of 64, a learning rate of 1×10^{-3} , and parameters set to $\text{betas} = (0.9, 0.999)$ and $\text{eps} = 1 \times 10^{-8}$. The model was fine-tuned for 30 epochs using DFA with the Adam optimizer and a learning rate of 1×10^{-3} .

4) Result Comparison: For the detection task results presented in Table V, we evaluated four training schemes, mirroring those used in the seizure prediction task. Due to the limited availability of ictal stage data, sensitivity, specificity, and F1 scores were not reported. Instead, the evaluation focused on accuracy-related measures to assess model performance. The pretrained model, trained exclusively on the first session, demonstrated an accuracy of 85.06%, indicating relatively good performance. This outcome is expected, as ictal and non-ictal stages are often easily distinguishable, even based on simple features like signal amplitude, despite the extended time intervals between sessions. The direct trained model achieved the highest accuracy of 98.22%. Our proposed MEIGA method with DFA consistently surpassed the pretrained model, achieving an accuracy that was 2.73% higher than the MEIGA model using backpropagation. The MEIGA model with DFA utilized only 2,008 parameters, constituting 17.40% of the backbone model size, highlighting its lightweight design. Notably, for specific patients, such as 08 and 10, DFA demonstrated clear advantages over BP, further emphasizing its effectiveness and potential in certain scenarios. This underscores the adaptability and efficiency of the DFA-based approach, particularly for edge-device applications where computational resources are constrained.

C. Ablation Study

1) Ablation on Leave-One-Session-Out Paradigm: In our previous experiments, we adopted a practical protocol where models were pre-trained on the first session and tested or adapted

TABLE VI

RESULTS COMPARISON TO RECENT STATE-OF-THE-ART METHODS FOR SEIZURE PREDICTION UNDER THE LEAVE-ONE-SESSION-OUT PROTOCOL

Subject	Truong et al. [55]			Lawhern et al. [57]			Daoud et al. [24]			Xu et al. [58]			Zhang et al. [59]			Wang et al. [60]			This Work		
	ACC	SS	SP	ACC	SS	SP	ACC	SS	SP	ACC	SS	SP	ACC	SS	SP	ACC	SS	SP	ACC	SS	SP
01	0.894	1	0.788	0.910	1	0.821	0.945	1	0.891	0.885	0.991	0.778	0.939	0.997	0.882	0.972	1	0.944	0.926	0.944	0.917
08	0.488	0.038	0.938	0.558	0.874	0.242	0.544	0.890	0.197	0.742	0.923	0.283	0.740	0.970	0.103	0.492	0.421	0.562	0.815	0.778	0.889
09	0.728	0.852	0.605	0.618	0.451	0.786	0.625	0.549	0.702	0.697	0.513	0.882	0.734	0.612	0.856	0.774	0.749	0.799	0.870	0.833	0.889
10	0.809	0.815	0.802	0.712	0.571	0.852	0.735	0.747	0.724	0.731	0.555	0.908	0.935	0.943	0.926	0.915	0.986	0.844	0.926	0.833	0.972
Average	0.730	0.676	0.783	0.700	0.724	0.675	0.712	0.797	0.629	0.764	0.746	0.713	0.837	0.881	0.692	0.788	0.789	0.787	0.884	0.847	0.917

to data from the subsequent session (Session 2). This setup closely aligns with real-world applications. However, another commonly used paradigm, leave-one-session-out (LOSO), is often employed in research [60]. In this approach, models are trained on all but one session and tested on the left-out session. While this paradigm is less practical—assuming that future data can predict historical seizures—it provides a valuable benchmark for assessing performance.

To assess the versatility of the proposed MEIGA framework, we compared our seizure prediction results with six state-of-the-art methods [24], [55], [57], [58], [59], [60]. Notably, while other methods are trained on multiple sessions and tested on a left-out session, MEIGA is pre-trained exclusively on the first session and adapted to the second session, demonstrating its adaptability with minimal data. Feature extraction methods from the reference works were implemented according to their original descriptions. Specifically: Truong et al. [55] employs short-time Fourier transform (STFT); Zhang et al. [59] uses the Pearson correlation coefficient; Other methods, including ours, rely on raw EEG signals.

To ensure fair comparison across methods, sampling lengths and sliding window parameters were adjusted to ensure consistency: [24], [57], [60] use a 5-second sliding window without overlap. [55], [58], [59] use 30-second (8-second overlap), 8-second (2-second overlap), and 20-second (5-second overlap) windows, respectively. For all compared sessions, a seizure-free interval of at least four hours between consecutive seizures was maintained. This design minimizes overlap between postictal and preictal stages, which is critical both for focusing on predicting the first seizure in a cluster—the most clinically actionable scenario—and for avoiding optimistic biases, as seizures within a cluster are generally easier to predict [64].

Under these conditions, our MEIGA framework achieved the highest average accuracy, sensitivity, and specificity among all methods, showing significant improvements over the state-of-the-art. Furthermore, MEIGA requires only 3,908 parameters, the second-lowest among all methods compared, highlighting its efficiency and suitability for resource-constrained environments. These results emphasize the practical advantages and robust performance of our approach, even when benchmarked against less realistic but widely used paradigms, as summarized in Fig. VI and Fig. VII.

2) Ablation on Cross-Validation Paradigms: While the previous section focused on the LOSO protocol, another widely adopted line of research employs cross-validation strategies such as leave-one-out cross validation (LOOCV) or k-fold cross validation. In these approaches, models are trained on nearly all available data except for one (or several) held-out folds and then tested on the excluded portion. Compared with LOSO,

TABLE VII

COMPARISON OF FEATURE EXTRACTION METHODS, ARCHITECTURES, AND PARAMETERS FOR COMPARED APPROACHES

Models	Feature extraction	Architecture	Parameters
Truong et al. [55]	STFT Spectral Images	CNN	224.066×10^3
Lawhern et al. [57]	Raw data	CNN	2.384×10^3
Daoud et al. [24]	Raw data	DCAE + Bi-LSTM	47.574×10^3
Xu et al. [58]	Raw data	CNN	2.040×10^6
Zhang et al. [59]	Pearson Correlation Coefficient	CNN	75.010×10^3
Wang et al. [60]	Raw data	CNN + Bi-LSTM	6.274×10^3
This work	Raw data	MEIGA	3.908×10^3

TABLE VIII

RESULTS COMPARISON UNDER THE CROSS-VALIDATION BASED METHODS ON THE CHB-MIT DATASET

Methods	Features	Architecture	Data Division	ACC (%)	SS (%)
Jang et al. [61]	STFT	Diminished MobileNet	LOOCV	86.94	—
Xu et al. [62]	Raw EEG	DynFCNet	LOOCV	88.00	88.40
Shan et al. [63]	Raw EEG	SincNet-based CNN	4-fold CV	87.20	88.25
Ours	Raw EEG	MEIGA	Pretrain-Adapt	86.77	85.32

such protocols leverage more training data and can be regarded as optimistic benchmarks, serving as a stronger direct trained scenario compared to our setting.

To further situate MEIGA within this broader context, we compared our results with three representative and recently introduced methods for their novelty and their use of cross-validation. Shan et al. [63] introduced an interpretable SincNet-based architecture, integrating sinc convolution layers into spatiotemporal CNNs for seizure prediction. Using the SincShallowConvNet model, it achieved a mean accuracy of 87.2% and a sensitivity of 88.3% on the CHB-MIT dataset, with the goal of improving both performance and interpretability by reducing parameters and visualizing bandpass filter ranges. Xu et al. [62] proposed a dual-branch neural network called DynFCNet, which combines a Dynamic Graph Convolutional Network with a standard CNN to process multichannel EEG signals. The model achieved an accuracy of 88.0% and a recall of 88.4% on the CHB-MIT dataset. Its objective is to better capture the complexity of EEG by extracting both Euclidean and non-Euclidean features. Jang et al. [61] introduced Diminished MobileNet, a lightweight seizure prediction model that employs STFT images from single-channel EEG data. On the CHB-MIT dataset, it achieved a maximum accuracy of 90.72% and an average of 86.94% in the frontal region, identified as the most informative measurement location. With only 37,985 parameters (1.176% of MobileNet), it targets computational efficiency suitable for wearable and mobile devices.

As shown in Table VIII, although cross-validation paradigms generally report higher absolute accuracies, MEIGA remains competitive and offers complementary advantages: it adapts with minimal session data while maintaining a lightweight parameter footprint suitable for resource-constrained environments.

TABLE IX

ACCURACY (%) COMPARISON OF MEIGA USING BP AND DFA ACROSS DIFFERENT COMPRESSION FACTORS (r) ON THE CHB-MIT DATASET

Subject	BP r_4	DFA r_4	BP r_8	DFA r_8	BP r_{16}	DFA r_{16}
1	98.15	94.44	94.44	92.59	88.89	92.59
4	88.89	75.93	92.59	81.48	77.78	53.70
5	79.63	70.37	81.48	85.19	77.78	74.07
7	88.89	81.48	87.04	87.04	87.04	85.19
8	87.04	72.22	83.33	81.48	61.11	62.96
9	81.48	72.22	81.48	87.04	70.37	61.11
10	94.44	92.59	96.30	92.59	75.93	87.04
AVG	88.36	79.89	88.09	86.77	76.99	73.81

TABLE X

ROC AUC COMPARISON OF MEIGA USING BP AND DFA ACROSS DIFFERENT COMPRESSION FACTORS (r) ON THE AES DATASET

Subject	BP r_8	DFA r_8	BP r_{16}	DFA r_{16}	BP r_{32}	DFA r_{32}
01	0.857	0.796	0.860	0.718	0.860	0.748
02	1.000	0.985	0.997	0.919	0.975	0.940
03	0.999	0.998	0.999	0.998	0.999	0.997
04	1.000	1.000	1.000	0.955	1.000	0.910
05	0.961	0.982	0.956	0.940	0.956	0.962
AVG	0.963	0.952	0.962	0.906	0.958	0.911

3) Ablation on Intrinsic Dimension Compression Factor: We conducted an ablation study to investigate the impact of different adaptor module sizes on the performance of the seizure prediction model. The study for the CHB-MIT dataset, shown in Table IX, modified only the tuning method (BP and DFA) and compression factor ($r = 4, 8$, and 16), while keeping all other hyperparameters fixed. The selection of compression factors was based on the backbone encoder module, which contains 12,704 parameters. All ablated models used the same pre-trained models for each subject. The results revealed that for the BP algorithm, accuracy decreased as the compression factor increased. Although the average performance degradation was minimal—only 0.25% from a compression factor of 4 to 8—five out of seven subjects experienced a more than 10% drop when r increased from 8 to 16. This suggests that there is a limit to the compactness of the MEIGA adaptor, and excessive compression can impair performance. Interestingly, for the DFA method, performance peaked when the compression factor was 8. However, performance dropped significantly when r increased to 16, suggesting that while moderate compression factors can improve performance, overly compact adaptors may fail to capture important features.

We further conducted an ablation study on the AES dataset. As shown in Table IX, we varied the tuning method and selected compression factors of 8, 16, and 32 for the larger encoder module. The corresponding side networks accounted for 9.28%, 4.58%, and 2.32% of the whole backbone model, which contains 2,141,954 parameters. Notably, Dog5 uses a 15-channel input, resulting in a slightly different parameterization and ratio. We observed that BP-trained adaptors achieved more stable results than DFA-trained adaptors, being less sensitive to the proportion of newly trained parameters. DFA maintained strong performance at a compression factor of 8, even surpassing BP on Dog 5. However, at higher compression rates, DFA exhibited a sharper drop in ROC AUC. These results emphasize the importance

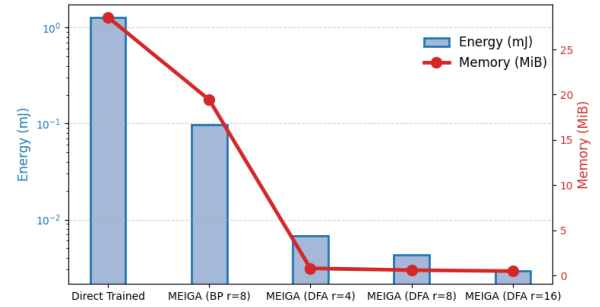


Fig. 9. Energy and Memory Comparison of Different Methods on the CHB-MIT dataset.

of selecting an optimal compression factor to strike a balance between efficiency and accuracy.

4) Energy and Memory Analysis: The primary benefit of MEIGA is fewer multiply-accumulate (MAC) operations and, consequently, lower compute energy. We quantify this on the CHB-MIT seizure prediction model for a single-sample backward pass using an operation-based energy model that multiplies MAC counts by FP32 per-operation costs (FAdd 0.9 pJ, FMult 3.7 pJ) [65]. We decompose the network into the embedding module, used only in the backbone, and the encoder module, which is present in both the backbone and the side net, and we exclude element-wise operations, activations, and dropout to focus on convolution, self-attention, and linear layers. Under DFA, the backward pass does not traverse the attention computation graph; no attention maps or value-weighted combinations are formed. Instead, output errors are projected directly to hidden layers via fixed feedback connections, while parameter-gradient computations for linear layers remain unchanged. In addition, the side net employs a compressed encoder with $r \in \{4, 8, 16\}$, further reducing MACs. As summarized in Fig. 9, for a single-sample backward pass the direct-trained backbone costs 1.2888 mJ. With the side net at $r=8$, BP costs 0.0976 mJ (7.6% of direct-trained), whereas DFA at $r=8$ costs 0.0043 mJ (0.33%), representing a 95.6% reduction compared to BP. Across DFA compression, energy decreases with larger r , from 0.0068 mJ at $r=4$ to 0.0029 mJ at $r=16$.

We also quantify activation memory required for a single-sample backward pass with FP32 tensors. This accounting includes the encoder inputs, Q, K, V projections, attention probabilities, and FFN intermediates. For the side net, it also includes the inputs to the five linear layers from the backbone encoder modules. We assume forward intermediates are cached in memory and no recomputation is used. Similarly, in DFA the backward pass does not traverse the attention graph; only the minimal encoder inputs and FFN intermediates are retained. As depicted in Fig. 9, with the side net at $r=8$, BP requires 19.44 MiB, whereas DFA uses 0.60 MiB (3.1% of BP). Within DFA, peak activation memory decreases with larger r , from 0.80 MiB at $r=4$ to 0.49 MiB at $r=16$.

V. CONCLUSION

This study introduces a novel framework named Memory-Efficient Intrinsic Gating Adaptation that integrates a

lightweight adaptor with Vision Transformer and CNN-based architectures, utilizing Direct Feedback Alignment to enhance the performance of on-device EEG-based epilepsy seizure detection and prediction. By combining pretraining with session-based tuning, the framework effectively addresses inefficiencies and limitations associated with sequential training in traditional methods, leveraging DFA's capability for parallel processing. A key strength of the proposed approach lies in its ability to adapt to the temporal variability of patient-specific biomarkers. Such variability often undermines the performance of pretrained models over time, yet this method ensures robust and consistent performance. Experimental results demonstrate that the framework achieves competitive accuracy in both detection and prediction tasks, rivaling state-of-the-art methods while maintaining a compact and efficient network architecture optimized for parallel tuning. The findings suggest that session-based tuning represents a practical and effective strategy for real-world seizure stage classification. This approach addresses the dynamic nature of patient biomarkers, enhancing model adaptability over time, and its adaptability and efficiency also make it promising for broader EEG-based health monitoring applications. Although this study focuses on epilepsy, MEIGA could potentially be extended to long-term sleep monitoring to deal with sleep-related neurological conditions [66].

REFERENCES

- I. Ahmad et al., "Robust epileptic seizure detection based on biomedical signals using an advanced multi-view deep feature learning approach," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 10, pp. 5742–5754, Oct. 2024.
- S. Zhao, J. Yang, and M. Sawan, "Energy-efficient neural network for epileptic seizure prediction," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 1, pp. 401–411, Jan. 2022.
- I. Ahmad, X. Wang, D. Javeed, P. Kumar, O. W. Samuel, and S. Chen, "A hybrid deep learning approach for epileptic seizure detection in EEG signals," *IEEE J. Biomed. Health Inform.*, early access, Apr. 10, 2023, doi: [10.1109/JBHI.2023.3265983](https://doi.org/10.1109/JBHI.2023.3265983).
- F. Tian, J. Yang, S. Zhao, and M. Sawan, "NeuroCARE: A generic neuromorphic edge computing framework for healthcare applications," *Front. Neurosci.*, vol. 17, 2023, Art. no. 1093865.
- K. K. Patro et al., "SMARTSeiz: Deep learning with attention mechanism for accurate seizure recognition in IoT healthcare devices," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 7, pp. 3810–3818, Jul. 2024.
- D. Wu, J. Yang, and M. Sawan, "Bridging the gap between patient-specific and patient-independent seizure prediction via knowledge distillation," *J. Neural Eng.*, vol. 19, no. 3, 2022, Art. no. 036035.
- X. Zhang, X. Zhang, Q. Huang, and F. Chen, "A review of epilepsy detection and prediction methods based on EEG signal processing and deep learning," *Front. Neurosci.*, vol. 18, 2024, Art. no. 1468967.
- K. Rasheed et al., "Machine learning for predicting epileptic seizures using EEG signals: A review," *IEEE Rev. Biomed. Eng.*, vol. 14, pp. 139–155, 2021.
- A. R. Khouas, M. R. Bouadjene, H. Hacid, and S. Aryal, "Training machine learning models at the edge: A survey," 2024, arXiv:2403.02619.
- V. Jayaram, M. Alamgir, Y. Altun, B. Scholkopf, and M. Grosse-Wentrup, "Transfer learning in brain-computer interfaces," *IEEE Comput. Intell. Mag.*, vol. 11, no. 1, pp. 20–31, Feb. 2016.
- A. L. Goldberger et al., "Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- B. H. Brinkmann et al., "Crowdsourcing reproducible seizure forecasting in human and canine epilepsy," *Brain*, vol. 139, no. 6, pp. 1713–1722, 2016.
- R. S. Fisher et al., "Epileptic seizures and epilepsy: Definitions proposed by the international league against epilepsy (ILAE) and the international bureau for epilepsy (IBE)," *Epilepsia*, vol. 46, no. 4, pp. 470–472, 2005.
- D. Wu, S. Li, J. Yang, and M. Sawan, "Neuro-BERT: Rethinking masked autoencoding for self-supervised neurological pretraining," *IEEE J. Biomed. Health Inform.*, early access, Jun. 18, 2024, doi: [10.1109/JBHI.2024.3415959](https://doi.org/10.1109/JBHI.2024.3415959).
- J. Gotman, "Automatic recognition of epileptic seizures in the EEG," *Electroencephalogr. Clin. Neurophysiol.*, vol. 54, no. 5, pp. 530–540, 1982.
- Y. Park, L. Luo, K. K. Parhi, and T. Netoff, "Seizure prediction with spectral power of EEG using cost-sensitive support vector machines," *Epilepsia*, vol. 52, no. 10, pp. 1761–1770, 2011.
- L. Gagliano, E. B. Assi, D. K. Nguyen, S. Rihana, and M. Sawan, "Bilateral preictal signature of phase-amplitude coupling in canine epilepsy," *Epilepsy Res.*, vol. 139, pp. 123–128, 2018.
- U. R. Acharya, H. Fujita, V. K. Sudarshan, S. Bhat, and J. E. Koh, "Application of entropies for automated diagnosis of epilepsy using EEG signals: A review," *Knowl.-Based Syst.*, vol. 88, pp. 85–96, 2015.
- E. B. Assi, D. K. Nguyen, S. Rihana, and M. Sawan, "A functional-genetic scheme for seizure forecasting in canine epilepsy," *IEEE Trans. Biomed. Eng.*, vol. 65, no. 6, pp. 1339–1348, Jun. 2018.
- R. Sharma and R. B. Pachori, "Classification of epileptic seizures in EEG signals based on phase space representation of intrinsic mode functions," *Expert Syst. Appl.*, vol. 42, no. 3, pp. 1106–1117, 2015.
- M. Behnam and H. Pourghassem, "Real-time seizure prediction using RLS filtering and interpolated histogram feature based on hybrid optimization algorithm of Bayesian classifier and hunting search," *Comput. Methods Programs Biomed.*, vol. 132, pp. 115–136, 2016.
- D. Wu, Y. Shi, Z. Wang, J. Yang, and M. Sawan, "C 2 SP-Net: Joint compression and classification network for epilepsy seizure prediction," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 841–850, 2023.
- K. M. Tsiouris, V. C. Pezoulas, M. Zervakis, S. Konitsiotis, D. D. Koutsouris, and D. I. Fotiadis, "A long short-term memory deep learning network for the prediction of epileptic seizures using EEG signals," *Comput. Biol. Med.*, vol. 99, pp. 24–37, 2018.
- H. Daoud and M. A. Bayoumi, "Efficient epileptic seizure prediction based on deep learning," *IEEE Trans. Biomed. Circuits Syst.*, vol. 13, no. 5, pp. 804–813, Oct. 2019.
- A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.
- R. V. Godoy et al., "EEG-based epileptic seizure prediction using temporal multi-channel transformers," 2022, arXiv:2209.11172.
- X. Xu, Y. Zhang, R. Zhang, and T. Xu, "Patient-specific method for predicting epileptic seizures based on DRNN-GRU," *Biomed. Signal Process. Control*, vol. 81, 2023, Art. no. 104449.
- M. Zhang, L. Zhang, C.-W. Tsai, and J. Yoo, "A patient-specific closed-loop epilepsy management SoC with one-shot learning and online tuning," *IEEE J. Solid-State Circuits*, vol. 57, no. 4, pp. 1049–1060, Apr. 2022.
- Z. Yuan, D. Zhang, Y. Yang, J. Chen, and Y. Li, "PPI: Pretraining brain signal model for patient-independent seizure detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, vol. 36, pp. 69586–69604.
- T. Dissanayake, T. Fernando, S. Denman, S. Sridharan, and C. Fookes, "Patient-independent epileptic seizure prediction using deep learning models," 2020, arXiv:2011.09581.
- B. Zhang et al., "Cross-subject seizure detection in EEGs using deep transfer learning," *Comput. Math. Methods Med.*, vol. 2020, no. 1, 2020, Art. no. 7902072.
- X. Cui, T. Wang, X. Lai, T. Jiang, F. Gao, and J. Cao, "Cross-subject seizure detection by joint-probability-discrepancy-based domain adaptation," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 2507413.
- V. Losing, B. Hammer, and H. Wersing, "Incremental on-line learning: A review and comparison of state of the art algorithms," *Neurocomputing*, vol. 275, pp. 1261–1274, 2018.
- S. C. Hoi, D. Sahoo, J. Lu, and P. Zhao, "Online learning: A comprehensive survey," *Neurocomputing*, vol. 459, pp. 249–289, 2021.
- S. Zhao et al., "A 0.99-to-4.38Uj/class event-driven hybrid neural network processor for full-spectrum neural signal analyses," *IEEE Trans. Biomed. Circuits Syst.*, vol. 17, no. 3, pp. 598–609, Jun. 2023.
- Y. Jewajinda and P. Chongstitvatana, "FPGA-based online-learning using parallel genetic algorithm and neural network for ECG signal classification," in *Proc. ECTI Int. Conf. Elect. Eng./Electron., Comput., Telecommun. Inf. Technol.*, 2010, pp. 1050–1054.
- A. Chua, M. I. Jordan, and R. Muller, "SOUL: An energy-efficient unsupervised online learning seizure detection classifier," *IEEE J. Solid-State Circuits*, vol. 57, no. 8, pp. 2532–2544, Aug. 2022.

- [38] S. Benatti, F. Montagna, V. Kartsch, A. Rahimi, D. Rossi, and L. Benini, "Online learning and classification of EMG-based gestures on a parallel ultra-low power platform using hyperdimensional computing," *IEEE Trans. Biomed. Circuits Syst.*, vol. 13, no. 3, pp. 516–528, Jun. 2019.
- [39] I. M. Guerra, D. Y. Barsakcioglu, and D. Farina, "Adaptive EMG decomposition in dynamic conditions based on online learning metrics with tunable hyperparameters," *J. Neural Eng.*, vol. 21, no. 4, 2024, Art. no. 046023.
- [40] D. Kiyasseh, T. Zhu, and D. Clifton, "A clinical deep learning framework for continually learning from cardiac signals across diseases, time, modalities, and institutions," *Nature Commun.*, vol. 12, 2021, Art. no. 4221.
- [41] A. Shahbazinia, F. Ponzina, J. A. Miranda Calero, J. Dan, G. Ansaloni, and D. A. Alonso, "Resource-efficient continual learning for personalized online seizure detection," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2024, pp. 1–7.
- [42] H. Zhang, D. Wu, R. Zhang, and R. Wang, "Incremental learning system for disease prediction: Epileptic seizure case," in *Proc. IEEE Glob. Commun. Conf.*, 2018, pp. 1–6.
- [43] Z. Deng, T. Mao, C. Shao, C. Li, and X. Chen, "Domain incremental learning for EEG-based seizure prediction," in *Proc. CAAI Int. Conf. Artif. Intell.*, 2023, pp. 476–487.
- [44] X. Wang, C. Zhang, T. Kärkkäinen, Z. Chang, and F. Cong, "Channel increment strategy-based 1D convolutional neural networks for seizure prediction using intracranial EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 316–325, 2023.
- [45] Y.-L. Sung, J. Cho, and M. Bansal, "LST: Ladder side-tuning for parameter and memory efficient transfer learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 12991–13005.
- [46] Z. Huo et al., "Decoupled parallel backpropagation with convergence guarantee," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 2098–2106.
- [47] C. Wolters, B. Taylor, E. Hanson, X. Yang, U. Schlichtmann, and Y. Chen, "Biologically plausible learning on neuromorphic hardware architectures," in *Proc. IEEE 66th Int. Midwest Symp. Circuits Syst.*, 2023, pp. 733–737.
- [48] J. Launay, I. Poli, and F. Krzakala, "Principled training of neural networks with direct feedback alignment," 2019, arXiv:1906.04554.
- [49] B. Crafton, A. Parihar, E. Gebhardt, and A. Raychowdhury, "Direct feedback alignment with sparse connections for local learning," *Front. Neurosci.*, vol. 13, 2019, Art. no. 525.
- [50] D. Han and H.-j. Yoo, "Efficient convolutional neural network training with direct feedback alignment," 2019, arXiv:1901.01986.
- [51] J. Launay, I. Poli, F. Boniface, and F. Krzakala, "Direct feedback alignment scales to modern deep learning tasks and architectures," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 9346–9360.
- [52] M. Nasser et al., "Ambulatory seizure forecasting with a wrist-worn device using long-short term memory deep learning," *Sci. Rep.*, vol. 11, no. 1, 2021, Art. no. 21935.
- [53] A. Antoniadis, L. Spyrou, C. C. Took, and S. Sanei, "Deep learning for epileptic intracranial EEG data," in *Proc. IEEE 26th Int. Workshop Mach. Learn. Signal Process.*, 2016, pp. 1–6.
- [54] E. B. Assi, M. Sawan, D. K. Nguyen, and S. Rihana, "A hybrid mRMR-genetic based selection method for the prediction of epileptic seizures," in *Proc. IEEE Biomed. Circuits Syst. Conf.*, 2015, pp. 1–4.
- [55] N. D. Truong et al., "Convolutional neural networks for seizure prediction using intracranial and scalp electroencephalogram," *Neural Netw.*, vol. 105, pp. 104–111, 2018.
- [56] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, arXiv:2010.11929.
- [57] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, 2018, Art. no. 056013.
- [58] Y. Xu, J. Yang, S. Zhao, H. Wu, and M. Sawan, "An end-to-end deep learning approach for epileptic seizure prediction," in *Proc. 2nd IEEE Int. Conf. Artif. Intell. Circuits Syst.*, 2020, pp. 266–270.
- [59] S. Zhang, D. Chen, R. Ranjan, H. Ke, Y. Tang, and A. Y. Zomaya, "A lightweight solution to epileptic seizure prediction based on EEG synchronization measurement," *J. Supercomputing*, vol. 77, pp. 3914–3932, 2021.
- [60] Z. Wang, J. Yang, H. Wu, J. Zhu, and M. Sawan, "Power efficient refined seizure prediction algorithm based on an enhanced benchmarking," *Sci. Rep.*, vol. 11, no. 1, 2021, Art. no. 23498.
- [61] D. Jang, K.-Y. Jung, Y.-G. Jeon, T.-J. Kim, S. K. Lee, and K. Min, "Diminished mobilenet: A lightweight architecture for epileptic seizure prediction using single-channel EEG," Available at SSRN 5097314, 2025.
- [62] T. Xu, Y. Wu, Y. Tang, W. Zhang, and Z. Cui, "Dynamic functional connectivity neural network for epileptic seizure prediction using multi-channel EEG signal," *IEEE Signal Process. Lett.*, vol. 31, pp. 1499–1503, 2024.
- [63] B. Shan, H. Yu, H. Jiang, Y. Huang, M. Xu, and D. Ming, "Interpretable sincnet-based spatiotemporal neural network for seizure prediction," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2024, pp. 1–6.
- [64] H.-H. Chen and V. Cherkassky, "Performance metrics for online seizure prediction," *Neural Netw.*, vol. 128, pp. 22–32, 2020.
- [65] M. Horowitz, "1.1 computing's energy problem (and what we can do about it)," in *Proc. IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, 2014, pp. 10–14.
- [66] T. Nakamura, V. Goverdovsky, M. J. Morrell, and D. P. Mandic, "Automatic sleep monitoring using ear-EEG," *IEEE J. Transl. Eng. Health Med.*, vol. 5, 2017, Art. no. 2800108.